Name ___________________________ Exam II-A Makeup Score ____________________

Section I. Short Answer Problems. Answer the problems on separate paper. You do <u>not</u> need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **<u>Show all relevant supporting steps!</u>** Attach this sheet to the front of your answers.

1. (5 pts) Find the determinant of the matrix A using an expansion by co-factors

where $A=\begin{bmatrix} 1 & -2 & 0 & 3 \\ 2 & -1 & -1 & 2 \\ 3 & -2 & -1 & 0 \\ 0 & 1 & -1 & 2 \end{bmatrix}$

2. (5 pts) Find the values of x for which the determinant of the matrix A is zero

where $A=\begin{bmatrix} 1 & x & 0 & -1 \\ 0 & -1 & x & 2 \\ x & 0 & 1 & -2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

3. (5 pts) Find the determinant of the matrix A by using elementary row operations to reduce A to a triangular matrix $A=\begin{bmatrix} 2 & -1 & 2 & 3 \\ -2 & 1 & 0 & -1 \\ 3 & -2 & 1 & 0 \\ 0 & 1 & -2 & 6 \end{bmatrix}$

4. (5 pts) Give an example to show that the determinant of a sum of matrices is, in general, not equal to the sum of the determinants of the matrices.

5. (5 pts) Determine whether the matrix A is singular where $A=\begin{bmatrix} -1 & 2 & 0 & 3 \\ 2 & 3 & 1 & 0 \\ -1 & 1 & -2 & 4 \\ 2 & 0 & -2 & 1 \end{bmatrix}$

6. (6 pts) Let $A=\begin{bmatrix} -2 & 1 & 0 & 2 \\ 1 & 4 & -1 & 2 \\ 3 & -1 & -2 & 1 \\ 0 & 2 & 1 & 2 \end{bmatrix}$. Find the determinant of: (a) A^T (b) A^5 (c) 6A

7. (6 pts) Let **u** = (2,2,-1,1), **v** = (-1,-3,-2,1), **w** = (2,-1,2,3).

Find: (a) 3**u** - 4**v** - 6**w** (b) **2v** + 3**u** - 4**w**

Section II. Answer these problems on these test pages.

8. (3 pts) For each of the following sets answer the question: Is this set with its specified operations a vector space? If the answer is no, give a reason for why the answer is no.

A. S = $\{p(x) \in P \mid p(-1) = -p(1)\}$,
addition is standard polynomial addition,
scalar multiplication is standard scalar multiplication for polynomials

B. S = $\{(a,b) \mid a,b > 0\}$,
addition is defined by $(a,b) \oplus (c,d) = (ac,bd)$
scalar multiplication is defined by $\alpha(a,b) = (e^{\alpha}a, e^{\alpha}b)$

C. S = $\left\{ \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & ab \end{bmatrix} \mid a,b \in \mathbb{R} \right\}$,
addition is standard matrix addition,
scalar multiplication is standard scalar multiplication

A. Yes No ____________________
B. Yes No ____________________
C. Yes No ____________________

9. (6 pts) For each of the following sets answer the question: Is this subset of P_3 a subspace of P_3? If the answer is no, give a reason for why the answer is no.

A. { p(t) $\in P_3$ | p(t) * p(-t) = 0 for all t }

B. { p(t) $\in P_3$ | $\int_{-\infty}^{\infty}$ p(t)dt = 0 }

C. { p(t) $\in P_3$ | p(1) *p(-1) = 0 }

D. { p(t) $\in P_3$ | p(1) *p(-1) = 1 }

E. { p(t) $\in P_3$ | p(0) = p(1)=p(2) }

F. { p(t) $\in P_3$ | p'(t) = t* p"(t) }

A. Yes No ____________________
B. Yes No ____________________
C. Yes No ____________________
D. Yes No ____________________
E. Yes No ____________________
F. Yes No ____________________

10. (6 pts) For each of the following sets answer the question: Is this set a spanning set for the given vector space?

A. { (3,5), (-6,–4) } for $\mathbb{R}^2$	Yes	No
B. { (1,-2), (-1,2), (-5,2) }, $\mathbb{R}^2$	Yes	No
C. { (-2,-1,-1), (2,5,-3), (-2,-4,2) }, $\mathbb{R}^3$	Yes	No
D. { (-2,1,2), (2,-1,1), (2,-1,3) }, $\mathbb{R}^3$	Yes	No
E. $\{x-1, x^2-1, x^2+1\}$, P_2	Yes	No
F. $\{x-1, -x^2+x-1, x^2+x-1\}$, P_2	Yes	No

11. (6 pts) For each of the following sets answer the question: Is this set a linearly independent subset of the given vector space?

A. { (3,5), (-6,–4) } for $\mathbb{R}^2$	Yes	No
B. { (1,-2), (-1,2), (-5,2) }, $\mathbb{R}^2$	Yes	No
C. { (-2,-1,-1), (2,5,-3), (-2,-4,2) }, $\mathbb{R}^3$	Yes	No
D. { (-2,1,2), (2,-1,1), (2,-1,3) }, $\mathbb{R}^3$	Yes	No
E. $\{x-1, x^2-1, x^2+1\}$, P_2	Yes	No
F. $\{x-1, -x^2+x-1, x^2+x-1\}$, P_2	Yes	No

12. (6 pts) For each of the following sets answer the question: Is this set a basis for the given vector space? If the answer is no, give a reason for why the answer is no.

A. { (3,5), (-6,–4) } for $\mathbb{R}^2$
B. { (1,-2), (-1,2), (-5,2) }, $\mathbb{R}^2$
C. { (-2,-1,-1), (2,5,-3), (-2,-4,2) }, $\mathbb{R}^3$
D. { (-2,1,2), (2,-1,1), (2,-1,3) }, $\mathbb{R}^3$
E. $\{x-1, x^2-1, x^2+1\}$, P_2
F. $\{x-1, -x^2+x-1, x^2+x-1\}$, P_2

A.	Yes	No	______________________________
B.	Yes	No	______________________________
C.	Yes	No	______________________________
D.	Yes	No	______________________________
E.	Yes	No	______________________________
F.	Yes	No	______________________________